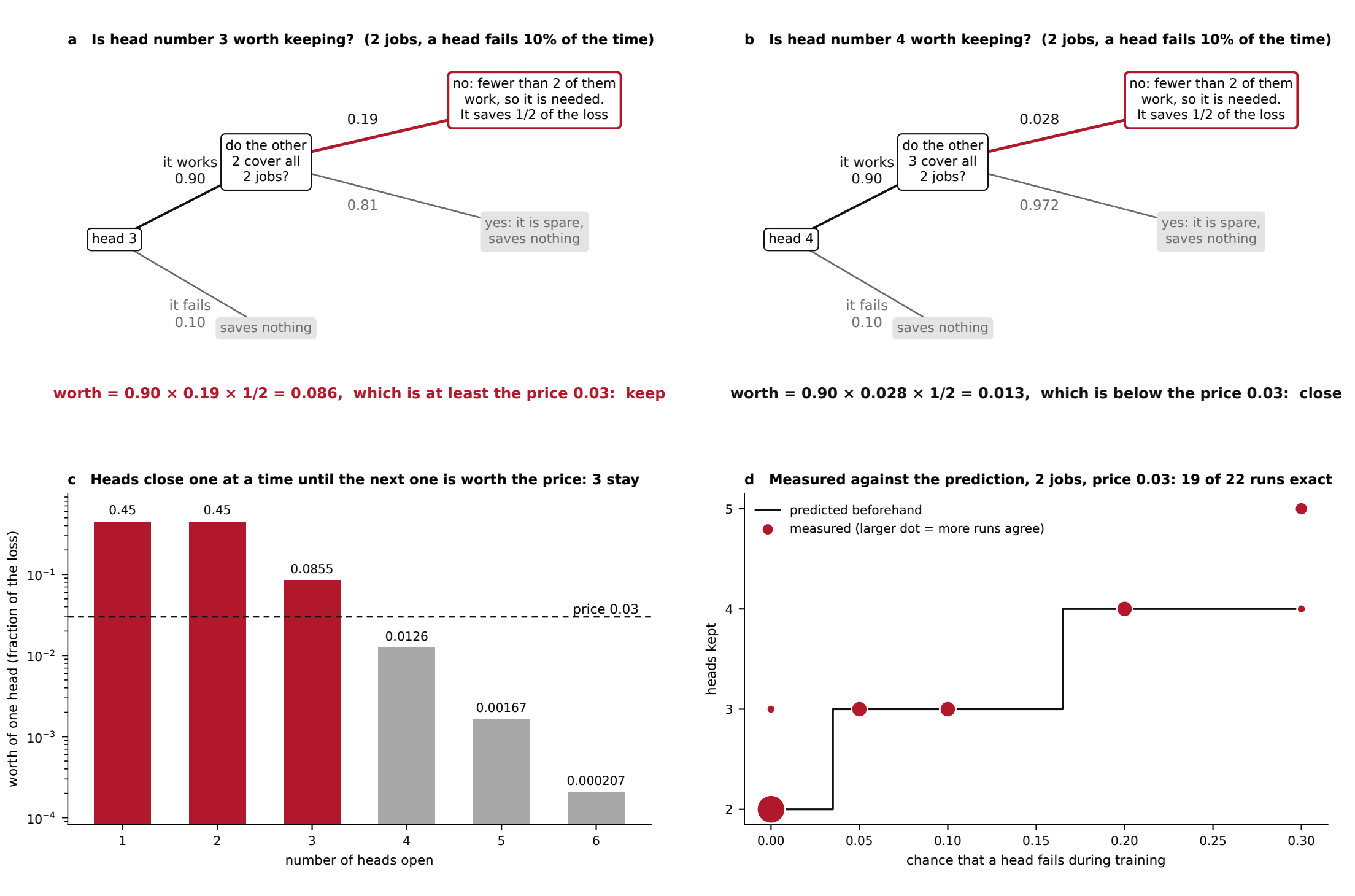
a Is head number 3 worth keeping? (2 jobs, a head fails 10% of the time)
head 3
it works
0.90
do the other
2 cover all
2 jobs?
0.19
no: fewer than 2 of them
work, so it is needed.
It saves 1/2 of the loss
0.81
yes: it is spare,
saves nothing
it fails
0.10
saves nothing
worth = 0.90 × 0.19 × 1/2 = 0.086, which is at least the price 0.03: keep
b Is head number 4 worth keeping? (2 jobs, a head fails 10% of the time)
head 4
it works
0.90
do the other
3 cover all
2 jobs?
0.028
no: fewer than 2 of them
work, so it is needed.
It saves 1/2 of the loss
0.972
yes: it is spare,
saves nothing
it fails
0.10
saves nothing
worth = 0.90 × 0.028 × 1/2 = 0.013, which is below the price 0.03: close
c Heads close one at a time until the next one is worth the price: 3 stay
worth of one head (fraction of the loss)
0.45
0.45
0.0855
0.0126
0.00167
0.000207
price 0.03
number of heads open
d Measured against the prediction, 2 jobs, price 0.03: 19 of 22 runs exact
predicted beforehand
measured (larger dot = more runs agree)
heads kept
chance that a head fails during training